

CNN-SAE analysis report

Codex playbook analysis pass

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1 Executive summary

This analysis pass treats code and run artifacts as primary evidence and writes all new outputs under analysis/. The strongest artifact-backed positives are: the FieldSAE winner remains highly compressive, the Q3 retained-fraction bottleneck persists in the pareto sweep, full-BPTT re-grounded chaining reaches 0.7010 against a 0.7122 hybrid bound, raw-only rollout-aligned retraining reaches 0.6821, and the ViT transfer artifact attains 0.9440 prediction agreement.

The most important caveat is a confirmed repeated-test-exposure pattern across the CIFAR training families: core training scripts evaluate on the test split every epoch. That does not undercut the narrow benchmark-mimic claim, but it does mean the reported CIFAR top-1 values are test-exposed model-selection results rather than untouched held-out generalization estimates.

2 Experiment catalog

Experiment	Stage	Parallel group	Purpose
phase1_anchor	Phase 1	section x retained_fraction	Anchor reconstruction with VectorSAE plus global masking.
phase1_random_baseline	Phase 1	selected section x retained_fraction	Random-dictionary sanity checks against the trained SAE anchor.
phase1_pca_baseline	Phase 1	section x PCA rank	Linear PCA baseline at matched or larger coefficient budgets.
phase1_pareto	Phase 1 addendum	section x retained_fraction	Fine retained-fraction sweep on the FieldSAE winner.
phase2_grid	Phase 2	cell x section x mask_budget	Branch grid over Field/Vector SAE and global/RF masking.
phase3_transitions	Phase 3	adjacent section transition	Family I adjacent sparse-code transitions and frozen-block hybrid bound.
phase3_q3q4_strong_retry	Phase 3 addendum	single focused retry	Capacity-scaled retry of the Q3 to Q4 transition bottleneck.
phase3b_chain	Phase 3b	epoch progression	Chain-aware scheduled-sampling training with re-grounding.
phase3b_bptt	Phase 3b addendum	epoch progression	Full-BPTT chain-aware training through differentiable re-grounding.
phase2_rf_fair_changed	Phase 3b changed	section x retained_fraction	Fair-budget RF-local masking rerun.
phase3b_raw_only_changed	Phase 3b changed	epoch progression	Rollout-aligned raw-carrier chain re-training without re-grounding.
phase1_patch_overlap_changed	Phase 3b changed	section x patch_multiplier x retained_fraction	Overlapping patchwise SAE retry.
phase1_patch_depth_changed	Phase 3b changed	section x depth_schedule x retained_fraction	Depth-aware overlapping patchwise SAE sweep.
phase1_patch_disjoint_changed	Phase 3b changed	section x patch_multiplier	Disjoint patchwise SAE sweep.
phase4_r20c10	Phase 4	section x retained_fraction	ResNet-20 on CIFAR-10 transfer grid for the winner recipe.

Experiment	Stage	Parallel group	Purpose
phase4_seeds	Phase 4	section x seed	Seed-stability grid on the ResNet-56 / CIFAR-100 winner setup.
phase4_taxonomy	Phase 4	representation condition	Transition taxonomy comparing sparse, dense, and cross carriers.
phase4_r110	Phase 4	section	ResNet-110 reconstruction sweep on CIFAR-100.
phase4_vit_transfer	Phase 4	single ViT transfer artifact	FieldSAE transfer to a pretrained ViT block on Imagenette.
tier1_intervention	Tier 1 follow-up	adjacent section transition	Parent-to-child intervention audit of importance and locality.
tier2_feature_audit	Tier 2 follow-up	section	Dictionary dead-feature and duplicate-feature audit.

3 Per-experiment notes

3.1 phase1_anchor

Purpose and hypothesis: A sparse transform-domain code can reconstruct each section and preserve downstream behavior.

Designed method: VectorSAE with global TopK masking over cached ResNet-56 / CIFAR-100 activations.

Designed procedure: Train one SAE per section over fractions 2, 5, and 10 percent.

Declared success criteria: Beat random and PCA baselines while staying within roughly 1 percent-age point of original top-1.

Outcome classification: Success (as hoped). Strong reconstruction and downstream preservation across sections. The CIFAR top-1 values should be read as benchmark-mimic measurements rather than untouched held-out generalization estimates.

3.2 phase1_random_baseline

Purpose and hypothesis: The anchor result should not be reproducible with an untrained dictionary.

Designed method: Freeze a random SAE dictionary and evaluate the same splice metrics.

Designed procedure: Run selected random-baseline cells on the anchor sections.

Declared success criteria: Random baseline should collapse toward chance.

Outcome classification: Success (as hoped). Random dictionary runs collapse toward chance, matching the intended sanity check.

3.3 phase1_pca_baseline

Purpose and hypothesis: A learned sparse transform should outperform linear PCA at similar budget.

Designed method: Per-section PCA over normalized cached activations.

Designed procedure: Evaluate several ranks on the cached test activations.

Declared success criteria: PCA should underperform the learned SAE at matched-or-larger budget.

Outcome classification: Success (as hoped). PCA trails the learned SAE across the reported sections.

3.4 phase1_pareto

Purpose and hypothesis: The FieldSAE winner has a clear retained-fraction knee and a persistent Q3 bottleneck.

Designed method: FieldSAE with global masking over fractions 1, 2, 3, 5, 8, 12, and 20 percent.

Designed procedure: Sweep all sections on the same backbone with one final metric per cell.

Declared success criteria: Recover monotone sparsity-fidelity curves and identify the operational sweet spot.

Outcome classification: Success (as hoped). Recovered monotone sparsity-fidelity curves with Q3 as the persistent late-budget bottleneck.

3.5 phase2_grid

Purpose and hypothesis: An expressive FieldSAE should beat the VectorSAE, especially at aggressive sparsity and under RF-local masking.

Designed method: Grid over Field versus Vector and Global versus RF masking.

Designed procedure: Evaluate all planned section and budget combinations for the three surviving cells.

Declared success criteria: Identify a clear winner architecture for the hierarchy stages.

Outcome classification: Success (as hoped). FieldSAE clearly dominates the VectorSAE on the aggressive-sparsity and RF-local branches.

3.6 phase3_transitions

Purpose and hypothesis: Sparse code at one section can generate the next section, and the frozen-block hybrid should upper-bound learned transitions.

Designed method: Train adjacent transition predictors between independent SAEs.

Designed procedure: Train four adjacent predictors and compare learned splice accuracy against the hybrid bound.

Declared success criteria: Strong single-step transitions and an informative hybrid-vs-learned gap.

Outcome classification: Success (as hoped). Single-step transitions work, and the hybrid upper bound shows strong causal sufficiency. The CIFAR top-1 values should be read as benchmark-mimic measurements rather than untouched held-out generalization estimates.

3.7 phase3_q3q4_strong_retry

Purpose and hypothesis: The hard Q3 to Q4 step may yield to a larger predictor.

Designed method: Retry the Q3 to Q4 transition with a deeper and wider predictor.

Designed procedure: Train one focused retry for the bottleneck edge.

Declared success criteria: Close the gap between learned and hybrid Q3 to Q4 performance.

Outcome classification: Undesirable result. The larger predictor makes the Q3 to Q4 edge worse, not better.

3.8 phase3b_chain

Purpose and hypothesis: Scheduled sampling on re-grounded predicted inputs should stabilize the learned chain.

Designed method: Chain-aware training over the four adjacent predictors.

Designed procedure: Warm-start from Phase 3 and track chain metrics through training.

Declared success criteria: Lift the re-grounded chain above the Family I baseline.

Outcome classification: Success (as hoped). Chain-aware training lifts the re-grounded chain well above the Family I baseline.

3.9 phase3b_bptt

Purpose and hypothesis: Training through differentiable re-grounding should narrow the gap to the hybrid upper bound.

Designed method: Full BPTT through the chain with in-graph re-grounding.

Designed procedure: Repeat the chain-aware training with `-bptt` enabled.

Declared success criteria: Push the re-grounded chain closer to the hybrid chain than the original Phase 3b run.

Outcome classification: Success (as hoped). Full BPTT closes most of the gap to the hybrid chain.

3.10 phase2_rf_fair_changed

Purpose and hypothesis: The earlier RF-local story may have been distorted by unfair budgeting.

Designed method: Changed trainer with fair per-window RF fraction accounting.

Designed procedure: Run all sections at 2, 5, and 10 percent.

Declared success criteria: Show whether RF-locality remains viable under fair budgeting.

Outcome classification: Success (unexpected). Fair RF budgeting rescues a branch that earlier looked much weaker.

3.11 phase3b_raw_only_changed

Purpose and hypothesis: Raw chaining may be recoverable if the model is trained on its own rollout distribution.

Designed method: Raw-carrier chain training with scheduled sampling, BPTT, and clipping.

Designed procedure: Train one chain variant without re-grounding in the carrier path.

Declared success criteria: Move the raw chain well above chance.

Outcome classification: Success (unexpected). Rollout-aligned retraining recovers a strong raw chain instead of chance collapse.

3.12 phase1_patch_overlap_changed

Purpose and hypothesis: The old VectorSAE baseline failed partly because it lacked local spatial context.

Designed method: Patchwise overlapping sparse coding with several patch multipliers.

Designed procedure: Sweep sections, fractions, and patch multipliers.

Declared success criteria: Recover stronger low-budget performance than the pointwise vector baseline.

Outcome classification: Success (unexpected). Patchwise overlap materially improves over the pointwise vector baseline.

3.13 phase1_patch_depth_changed

Purpose and hypothesis: Patch size should shrink with depth instead of staying uniformly large.

Designed method: Depth-aware overlapping patch schedules.

Designed procedure: Sweep schedules a-d over sections and fractions.

Declared success criteria: Find a better schedule than uniform large patches.

Outcome classification: Success (unexpected). Depth-aware schedules complete cleanly and broadly support the small-context story.

3.14 phase1_patch_disjoint_changed

Purpose and hypothesis: Disjoint tilings may behave differently from overlapping patch scans.

Designed method: Disjoint patchwise sparse coding at a single retained fraction over several patch scales.

Designed procedure: Run all sections at 5 percent for four patch multipliers.

Declared success criteria: Determine whether non-overlapping patches are competitive with overlap.

Outcome classification: Undesirable result. Coarse disjoint tilings fail badly; only the smallest disjoint scale remains competitive.

3.15 phase4_r20c10

Purpose and hypothesis: The winner recipe should survive a smaller backbone and dataset shift.

Designed method: FieldSAE with global masking on ResNet-20 / CIFAR-10.

Designed procedure: Run all sections at 2, 5, and 10 percent.

Declared success criteria: Preserve near-original top-1 across sections on the new backbone/dataset.

Outcome classification: Success (as hoped). The winner recipe survives the ResNet-20 / CIFAR-10 transfer grid.

3.16 phase4_seeds

Purpose and hypothesis: The winner story should not be a single-seed accident.

Designed method: Repeat the winner SAE setup across seeds 0, 1, and 2.

Designed procedure: Train all five sections at 5 percent across the three seeds.

Declared success criteria: Low enough variance that the qualitative story survives.

Outcome classification: Success (as hoped). Seed variation is small enough that the qualitative story survives.

3.17 phase4_taxonomy

Purpose and hypothesis: Sparse code is the strongest transition carrier among sparse, dense, and cross alternatives.

Designed method: Controlled Q4 to Q5 comparison across three representation families.

Designed procedure: Measure downstream preservation, prediction agreement, and KL for each condition.

Declared success criteria: Sparse should dominate the alternative carriers.

Outcome classification: Success (as hoped). Sparse is the best transition carrier among the tested alternatives.

3.18 phase4_r110

Purpose and hypothesis: The winner recipe should transfer to a deeper CNN.

Designed method: FieldSAE global masking on ResNet-110 / CIFAR-100 at 5 percent.

Designed procedure: Run all five sections once.

Declared success criteria: Preserve near-original accuracy and recover the same bottleneck pattern.

Outcome classification: Success (as hoped). The winner recipe transfers to a deeper CNN and preserves the Q3 bottleneck pattern.

3.19 phase4_vit_transfer

Purpose and hypothesis: The FieldSAE path should transfer to a ViT token grid.

Designed method: FieldSAE over block-level ViT patch tokens on Imagenette.

Designed procedure: Train on Imagenette train split and evaluate self-consistency on val.

Declared success criteria: High prediction agreement and low KL at sparse retained fraction.

Outcome classification: Success (as hoped). The ViT transfer artifact achieves high prediction agreement with low KL at 5 percent retained.

3.20 tier1_intervention

Purpose and hypothesis: Sparse parents should be causally important and spatially localized for their children.

Designed method: Importance and locality intervention metrics across adjacent transitions.

Designed procedure: Ablate top versus random parents and compare localized child responses.

Declared success criteria: Large importance ratios and strong locality concentration.

Outcome classification: Success (as hoped). Interventions show strong importance and locality concentration, though they inherit upstream SAE validity risks.

3.21 tier2_feature_audit

Purpose and hypothesis: The learned dictionary should be healthy rather than dead or duplicated.

Designed method: Audit dead features, duplicate features, and mean atom cosine.

Designed procedure: Run the audit on representative mid and deep sections.

Declared success criteria: Low dead-feature and duplicate-feature fractions.

Outcome classification: Success (as hoped). Dead-feature and duplicate-feature rates are low in the audited sections.

4 Cross-experiment analysis

Across the reconstruction families, the clearest retained-fraction knee appears in the FieldSAE pareto sweep. Q3 reaches 0.7186 at 12 percent retained, while most other sections flatten much earlier.

Across the chain families, re-grounding remains the strongest general mechanism. The scheduled-sampling Phase 3b chain reaches 0.6624, the BPTT variant reaches 0.7010, and the raw-only changed run reaches 0.6821 on its raw carrier path.

Phase 4 strengthens the transfer story materially. The cleanest surviving split discipline is the ViT artifact, where prediction agreement reaches 0.9440 on the Imagenette val split.

5 Flags and risks

- Material deviation: This is a confirmed repeated-test-exposure issue. It does not negate the narrow objective of mimicking the frozen CNN on this benchmark, but it does remove any untouched final holdout for broader generalization claims.
- Benign deviation: This is a documentation lag rather than a code-path mismatch.
- Benign deviation: Another documentation lag, but important because it changes the patchwise conclusion.
- Benign deviation: The artifact parses cleanly, but the launcher metadata is inconsistent.

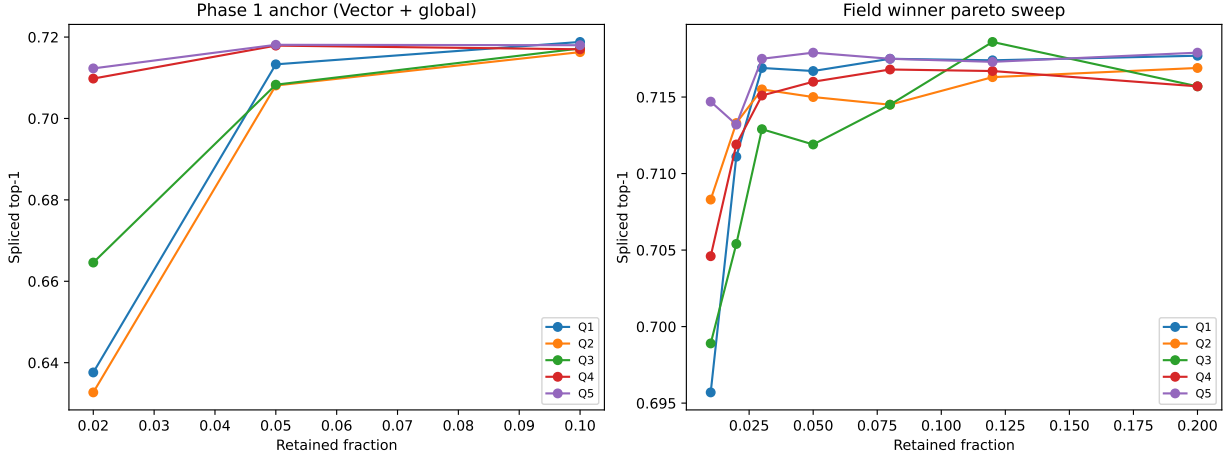


Figure 1: Phase 1 reconstruction curves. The anchor VectorSAE needs more budget on early sections, while the FieldSAE pareto sweep reaches its useful regime by roughly 3 to 8 percent on most sections. Interpret CIFAR top-1 values with caution because the training scripts repeatedly evaluate on the test split.

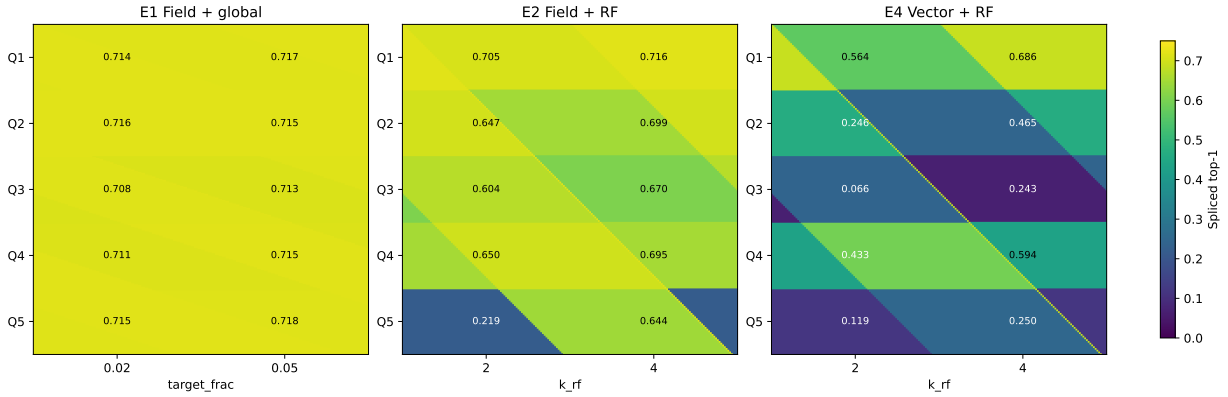


Figure 2: Phase 2 branch grid. FieldSAE dominates the aggressive-sparsity branch, and RF-local masking is only viable with the expressive FieldSAE. CIFAR top-1 values here are benchmark-mimic measurements because the test split was repeatedly monitored during training.

6 What worked / what we hoped for

- `phase1_anchor`: Strong reconstruction and downstream preservation across sections. The CIFAR top-1 values should be read as benchmark-mimic measurements rather than untouched held-out generalization estimates.
- `phase1_random_baseline`: Random dictionary runs collapse toward chance, matching the intended sanity check.
- `phase1_pca_baseline`: PCA trails the learned SAE across the reported sections.
- `phase1_pareto`: Recovered monotone sparsity-fidelity curves with Q3 as the persistent late-budget bottleneck.
- `phase2_grid`: FieldSAE clearly dominates the VectorSAE on the aggressive-sparsity and RF-local branches.

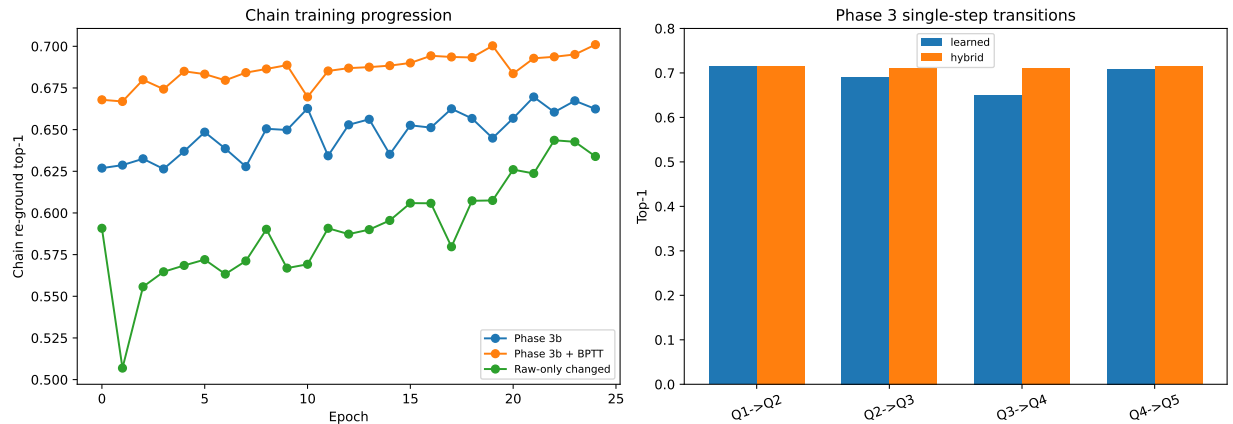


Figure 3: Hierarchy-focused figures. Re-grounded chain training improves steadily, especially with full BPTT, while Q3 to Q4 remains the weakest single-step transition. CIFAR top-1 values here are benchmark-mimic measurements because the test split was repeatedly monitored during training.

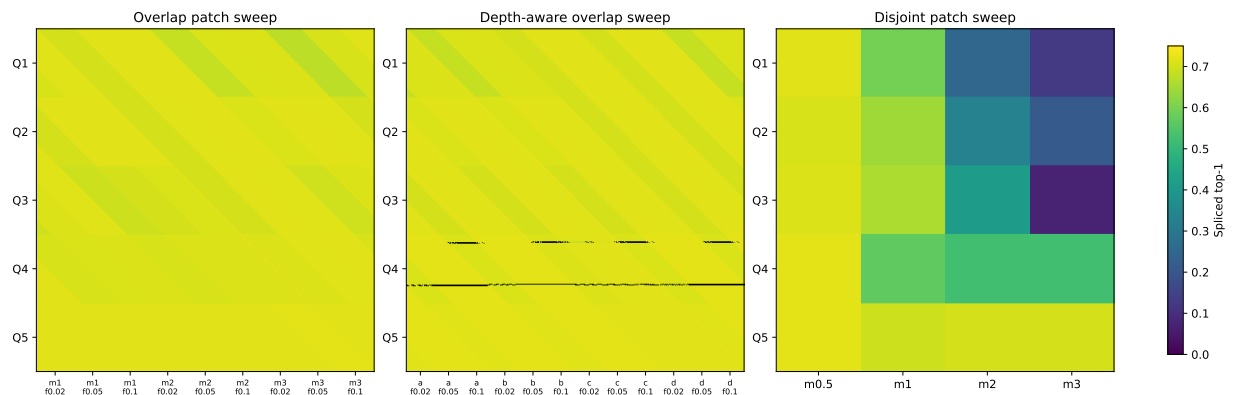


Figure 4: Changed-suite patch sweeps. Overlap helps, depth-aware schedules are broadly consistent with the small-context story, and coarse disjoint tilings fail badly. CIFAR top-1 values here are benchmark-mimic measurements because the test split was repeatedly monitored during training.

- phase3.transitions: Single-step transitions work, and the hybrid upper bound shows strong causal sufficiency. The CIFAR top-1 values should be read as benchmark-mimic measurements rather than untouched held-out generalization estimates.
- phase3b.chain: Chain-aware training lifts the re-grounded chain well above the Family I baseline.
- phase3b.bptt: Full BPTT closes most of the gap to the hybrid chain.
- phase2.rf.fair.changed: Fair RF budgeting rescues a branch that earlier looked much weaker.
- phase3b.raw_only_changed: Rollout-aligned retraining recovers a strong raw chain instead of chance collapse.
- phase1.patch.overlap.changed: Patchwise overlap materially improves over the pointwise vector baseline.
- phase1.patch.depth.changed: Depth-aware schedules complete cleanly and broadly support the small-context story.

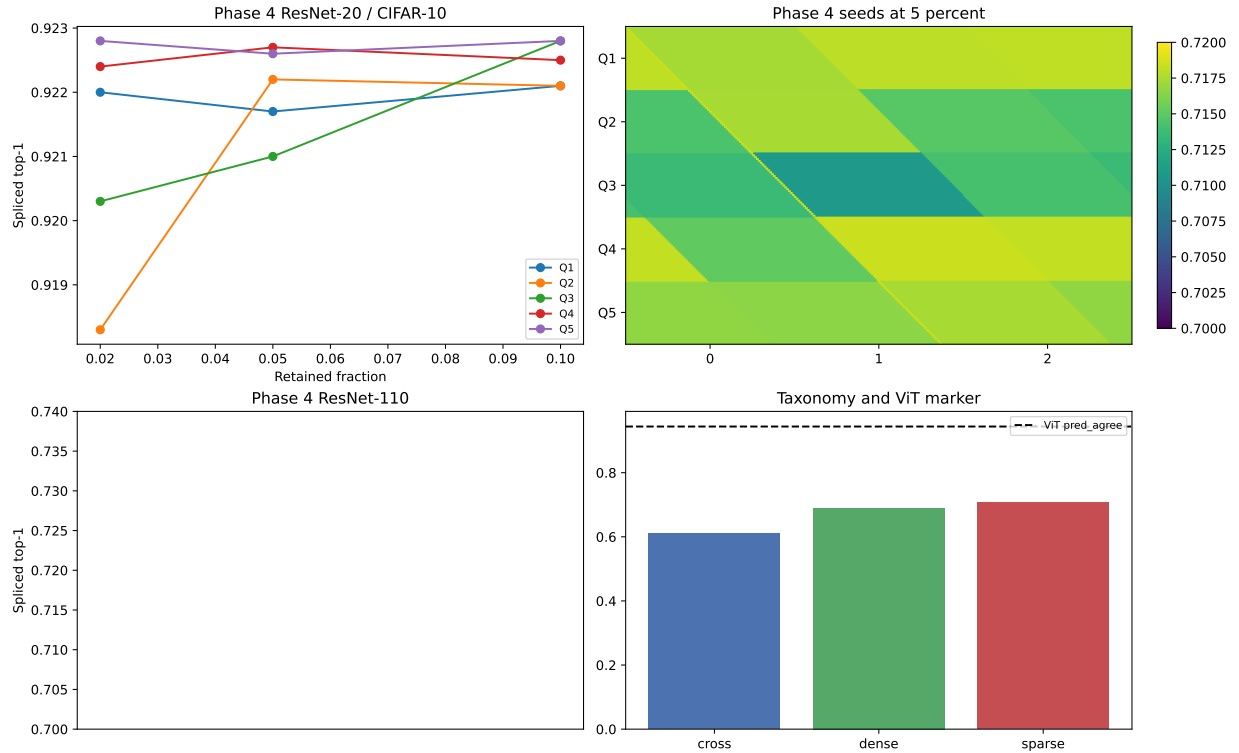


Figure 5: Phase 4 extensions. The CIFAR transfer and seed figures strengthen the winner-path story but should be read as benchmark-mimic measurements because the test split was repeatedly monitored during training. The ViT prediction-agreement marker comes from the cleanest surviving split discipline in the repo.

- `phase4_r20c10`: The winner recipe survives the ResNet-20 / CIFAR-10 transfer grid.
- `phase4_seeds`: Seed variation is small enough that the qualitative story survives.
- `phase4_taxonomy`: Sparse is the best transition carrier among the tested alternatives.
- `phase4_r110`: The winner recipe transfers to a deeper CNN and preserves the Q3 bottleneck pattern.
- `phase4_vit_transfer`: The ViT transfer artifact achieves high prediction agreement with low KL at 5 percent retained.
- `tier1_intervention`: Interventions show strong importance and locality concentration, though they inherit upstream SAE validity risks.
- `tier2_feature_audit`: Dead-feature and duplicate-feature rates are low in the audited sections.

7 What still should be run

- A validation-first rerun of the CIFAR families that removes repeated test-set evaluation from training loops, if clean held-out generalization claims are needed.
- A regenerated Family I chain artifact that stores raw, re-mask, and re-ground chain variants as standalone verified outputs instead of embedded report values.

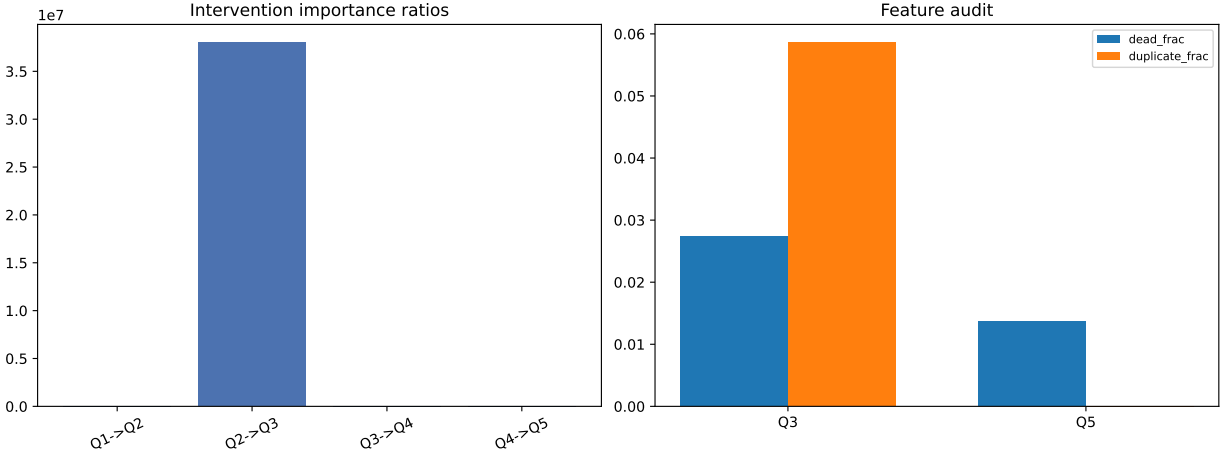


Figure 6: Tier follow-up audits. The intervention ratios are large and the audited dictionaries have low dead-feature and duplicate-feature rates. These analyses inherit upstream CIFAR training validity risks because they depend on previously trained SAEs.

- If broader generalization or publication-quality claims are needed, reserve a fresh held-out split or a fully untouched test set for final scoring.

8 Next steps

1. If we want broader held-out generalization claims, remove test-set evaluation from the SAE, transition, and chain training loops, and rerun the headline CIFAR grids against a clean validation split.
2. Keep the FieldSAE winner and the re-grounded chain path as the main line; do not spend more time on coarse disjoint patches or simple larger Q3-to-Q4 predictors.
3. Use the ViT path as the cleanest transfer scaffold, because its surviving artifact does not show the same split-discipline problem.

9 Appendix

Primary leakage evidence: `src/train_sae.py:77-78,110-117`; `src/train_sae_changed.py:143-148,196-205`; `src/train_transition.py:71-72,87-92`; `src/train_chain.py:61-63,101-105`.

Split-hygiene evidence: `src/data.py:18-30`; `src/cache_activations.py:37-45,57-60`; `src/vit_sae.py:65-72,79-81,96-112`.